PHYSICAL CLIMATE RISK: FLOOD MODEL COMPARISON, HAZARD AND VULNERABILTY ASSESSMENT AND FINANCIAL IMPLICATIONS

Klemen Kubelj

Master Thesis
Jožef Stefan International Postgraduate School
Ljubljana, Slovenia

Supervisor: Prof. Mitja Luštrek, Institut Jožef Stefan, Ljubljana, Slovenia
Co-Supervisor: Prof. Boštjan Kaluža, Arvio d.o.o, Ljubljana, Slovenia

Evaluation Board:
Prof. dr. Mitja Luštrek, Chair, Institut Jožef Stefan, Ljubljana, Slovenia
dr. Tea Tušar, Member, Institut Jožef Stefan, Ljubljana, Slovenia
dr. Inna Novalija, Member, Institut Jožef Stefan, Ljubljana, Slovenia

MEDNARODNA PODIPLOMSKA ŠOLA JOŽEFA STEFANA
JOŽEF STEFAN INTERNATIONAL POSTGRADUATE SCHOOL

Klemen Kubelj

PHYSICAL CLIMATE RISK: FLOOD MODEL COMPARISON, HAZARD AND VULNERABILTY ASSESSMENT AND FINANCIAL IMPLICATIONS

Master Thesis

NASLOV V SLOVENŠČINI

Magistrsko delo

Supervisor: Prof. Mitja Luštrek

Co-Supervisor: Prof. Boštjan Kaluža

Ljubljana, Slovenia, Month Year

Acknowledgments

Thank everyone who contributed to the thesis. Do not forget to mention any funding received.

Abstract

Flooding is one of the most frequent and damaging natural hazards globally, and its risk is projected to increase due to climate change. Accurate flood risk assessments are essential for spatial planning, climate adaptation, and ESG-aligned financial decision-making. This thesis investigates physical flood risk by comparing global and local hazard models, evaluating standardized vulnerability functions, and quantifying financial implications for real estate assets in Slovenia.

Two flood hazard models are assessed: the global WRI Aqueduct River Flood model and the high-resolution local model Integralna karta globin (IKG). A geospatial validation dataset comprising flood depth and damage records from four flood events in Slovenia (2010–2023) is used to evaluate model accuracy. Flood depths are analyzed using both regression and classification metrics, while vulnerability is assessed through comparison of observed damages against standardized depth–damage functions provided by the European Commission Joint Research Centre (JRC).

A key technical contribution is the development of a data processing pipeline that harmonizes heterogeneous IKG hazard data and integrates it into the modular Physrisk platform using the Zarr data format. This enables scalable, scenario-based risk assessment and model intercomparison.

Findings suggest that while local models offer improved accuracy for continuous flood depth prediction in regression tasks, the global model performs better in categorical classification. Proximity to water bodies was not found to significantly improve prediction accuracy. Depth–damage functions align well with observed damage distributions for residential properties, though segmentation by property type reveals variability in model performance.

The thesis contributes to the validation and integration of physical climate risk models for real estate applications and enables more accurate, location-specific flood risk estimates to support planning and investment decisions.

Povzetek

Povzetek v slovenščini naj ne bo daljši od ene strani.

Contents

List of Figures xiii

List of Tables xv

List of Algorithms xvii

Abbreviations xix

Symbols xxi

Glossary xxiii

1 Introduction 1
- 1.1 Background and Motivation . . . 1
- 1.2 Climate Risk and ESG Disclosure . . . 1
- 1.3 Problem Statement . . . 2
- 1.4 Research Objectives . . . 2
- 1.5 Structure of the Thesis . . . 3

2 Literature Review & Related Work 5

3 Study Area, Data, and Methodology 7
- 3.1 Hazard and Depth . . . 7
  - 3.1.1 Mathematical Definition of Flood Hazard . . . 7
  - 3.1.2 Hazard Data Sources and Modeling Approach . . . 8
- 3.2 Vulnerability and Damage Function . . . 9
  - 3.2.1 Mathematical Definition of Flood Vulnerability . . . 9
  - 3.2.2 Vulnerability Data Source . . . 9
  - 3.2.3 Validation Dataset . . . 10
  - 3.2.4 Assumptions and Limitations . . . 10
- 3.3 Flood Risk Modeling . . . 11
  - 3.3.1 Risk for a Single Return Period . . . 11
  - 3.3.2 Expected Annual Damage (EAD) . . . 11
  - 3.3.3 Exposure Data . . . 12
  - 3.3.4 Use Cases and Applications . . . 12

4 Model Integration and Analysis Framework 15
- 4.1 Architectural Diagram . . . 15
  - 4.1.1 System Overview . . . 16
  - 4.1.2 Data Flow and Model Interoperability . . . 16
  - 4.1.3 Integration of the IKG Model . . . 16
  - 4.1.4 Hazard Onboarding Pipeline . . . 16

4.2 Pipeline for Combining Heterogeneous IKG Models . . . 17
4.2.1 Data Sources and Heterogeneity . . . 17
4.2.2 Standardization and Processing Workflow . . . 17
4.2.3 Unified Flood Hazard Product . . . 18
4.3 Physrisk Framework – Model Onboarding . . . 18
4.3.1 Physrisk Architecture and Modules . . . 18
4.3.2 Zarr Format and Indexing . . . 19
4.3.3 Onboarding Procedure . . . 19
4.3.4 Integration with Other Models . . . 19
4.4 Comparison Toolset . . . 20
4.4.1 Purpose and Design Principles . . . 20
4.4.2 Interface and Functionality . . . 20
4.4.3 Supported Models and Inputs . . . 20
4.4.4 Integration with Validation Workflow . . . 21
4.4.5 Reproducibility and Extensibility . . . 21

5 Experimental Design and Evaluation 23
5.1 Experimental Goals . . . 23
5.2 Experimental Setup . . . 23
5.3 Experimental Results: Flood Depth . . . 23
5.4 Experimental Results: Damage Functions . . . 23
5.5 Analysis of Impact on Total Exposed Property Value . . . 23

6 Conclusions & Discussion 25
6.1 Conclusion & Interpretation of Results . . . 25
6.2 Limitations of This Work . . . 25
6.3 Future Work . . . 25

Appendix A Proofs of Theorems 27
A.1 Proof of the Pythagorean Theorem . . . 27

References 29

Bibliography 31

Biography 33

Index 35

List of Figures

Figure 3.1: Example hazard map and depth–return period curve. 8
Figure 3.2: Depth–damage function for residential buildings. 10
Figure 3.3: Expected Annual Damage from flood risk. 12

Figure A.1: Similar triangles used in the proof of the Pythagorean theorem. 27

List of Tables

Table 3.1: Comparison of flood hazard models used in the study. 9

List of Algorithms

Abbreviations

JSI ... Jožef Stefan Institute
IPS ... International Postgraduate School
AEP ... Average Exceedance Probability
EAD ... Expected Annual Damage
ESG ... Environmental, Social, and Governance
JRC ... Joint Research Centre
RP ... Return Period
WRI ... World Resources Institute

Symbols

$j^\star$ ... black-body irradiance
σ ... Stefan's (or Stefan-Boltzmann) constant

Glossary

Glossary of terms

Chapter 1

Introduction

1.1 Background and Motivation

Flooding is one of the most frequent and damaging natural hazards globally, responsible for extensive economic losses, social disruption, and loss of life. Climate change is expected to further intensify these risks by increasing both the frequency and severity of hydrological extremes. The Intergovernmental Panel on Climate Change (IPCC) projects a growing incidence of intense precipitation events and flood-related disasters, particularly in regions affected by higher hydrological variability and sea-level rise [1].

In Europe, recent catastrophic flood events—such as those in Germany and Belgium in 2021, and in Slovenia in 2023—have exposed significant vulnerabilities in infrastructure, spatial planning, and emergency preparedness. According to the European Environment Agency (EEA), riverine and pluvial floods already account for more than one-third of total weather-related losses in Europe, a share that is projected to increase over coming decades without improved mitigation and adaptation measures [2].

These developments underscore the urgent need for robust and spatially detailed flood risk models that can support risk-sensitive decision-making in domains such as insurance pricing, urban planning, asset valuation, and regulatory compliance. This is particularly critical for stakeholders in the real estate and financial sectors, who are increasingly required to assess and disclose exposure to climate-related physical risks as part of their environmental, social, and governance (ESG) reporting obligations [3], [4].

1.2 Climate Risk and ESG Disclosure

Climate risk refers to the potential adverse effects of climate change on economic systems, institutions, and physical assets. It is commonly classified into three interrelated categories: *physical*, *transition*, and *liability* risks [5].

- **Physical risks** arise from the direct consequences of climate variability and long-term climate change. These include acute hazards such as floods, storms, and wildfires, as well as chronic effects such as sea-level rise and shifting temperature trends. These risks can cause immediate physical damage and long-term degradation of asset value.

- **Transition risks** arise from the transition to a low-carbon economy. They include policy changes (e.g., carbon pricing, emissions regulations), technology disruption, evolving market preferences, and reputational risks associated with inaction or misalignment.

- **Liability risks** are considered by some [6] and encapsulate risks arising when those affected by anthoropogenic climate change seek compensation.

These categories are increasingly embedded in ESG-aligned regulatory frameworks such as those developed by the Task Force on Climate-related Financial Disclosures (TCFD) and the Network for Greening the Financial System (NGFS). In the European context, the European Central Bank (ECB) requires significant institutions to assess, manage, and disclose climate-related and environmental risks, including physical risks. According to the ECB's supervisory guidance, banks are expected to integrate these risks into their internal governance, risk management, stress testing, and disclosure practices [3].

Physical risk is typically modeled as the interaction of three fundamental components:

- **Hazard** – the probability and intensity of a climate-related event (e.g., flood depth for a 100-year return period),
- **Vulnerability** - the expected damage to an asset given the intensity of the hazard,
- **Exposure** - the presence and value of assets located in potentially affected areas.

These components form the foundation of physical climate risk models used in both regulatory compliance and voluntary disclosures. However, despite the wide adoption of this conceptual framework, practical implementation often suffers from limited model transparency, inconsistent spatial resolution, and poor validation—especially at the local scale relevant to property-level investment and risk management.

This thesis focuses specifically on **acute physical risks**, with a particular emphasis on riverine flooding and its financial impact on real estate assets. By comparing the performance of global and local flood models and evaluating standardized vulnerability functions against observed data, the work aims to contribute to more accurate and transparent risk quantification in support of ESG-aligned decision-making.

1.3 Problem Statement

While global flood hazard models such as the WRI Aqueduct River Floods provide broad coverage and climate scenario integration, they are often limited in spatial resolution and may not reflect local hydrological conditions. On the other hand, high-resolution local models, such as Slovenia's *Integralna karta globin* (IKG), offer detailed inundation mapping but are typically fragmented, static (i.e., not climate scenario-aware), and difficult to integrate into broader risk platforms.

Furthermore, flood vulnerability is typically modeled using generalized depth–damage functions derived from empirical or synthetic data. These functions may perform well in aggregate, but their applicability across different building types, geographies, and damage-reporting standards remains uncertain. This creates uncertainty in downstream financial estimates and limits the usefulness of such models for asset-level ESG disclosure.

1.4 Research Objectives

The purpose of this thesis is to evaluate how well existing flood models can represent real-world flood depths and damage outcomes in Slovenia. Specifically, it compares a global hazard model (WRI Aqueduct) and a high-resolution local model (IKG) to assess the accuracy of their flood depth predictions.

Separately, the thesis tests global depth-damage functions, used to estimate flood-related property value loss, against reported damage data, with the goal of evaluating and potentially improving their performance. Together, these analyses aim to assess the reliability of current modeling tools for understanding physical climate risk and supporting property related decision making in the context of increasing flood exposure.

The main goals of the research are:

1. To improve existing physrisk risk assessment framework with the following contributions:
   (a) Develop a data processing pipeline for creating a unified and homogenous SI IKG flood depth model for different areas of importance.
   (b) Develop a data onboarding pipeline for integrating this new unified model into physrisk framework.
2. To compare the performance of a global flood hazard model (WRI Aqueduct) and a local Slovenian model (IKG) using real flood data.
3. To evaluate how model resolution and proximity to water bodies affect prediction accuracy.
4. To assess the accuracy of global flood depth–damage functions on reported damage cases in Slovenia and improve their performance.

1.5 Structure of the Thesis

The remainder of the thesis is structured as follows:

- **Chapter 2** presents a review of relevant literature and related work on flood hazard modeling, vulnerability estimation, and ESG-aligned risk frameworks.
- **Chapter 3** describes the study area, datasets, and the methodological framework used for model evaluation and data processing.
- **Chapter 4** details the integration of the local IKG model into the Physrisk framework and presents the architecture of the comparison toolset.
- **Chapter 5** provides experimental results, including quantitative evaluation of model performance, depth–damage function analysis, and financial impact estimates.
- **Chapter 6** discusses the key findings, limitations, and possible future research directions, and concludes the thesis.

Chapter 2

Literature Review & Related Work

TBD

Chapter 3

Study Area, Data, and Methodology

This chapter introduces the geographic scope, data sources, and mathematical models used to evaluate flood hazard, vulnerability, and risk. The modeling framework is structured around three key components: **hazard** (depth), **vulnerability** (damage functions), and **exposure** (asset value). These elements are formally defined and then combined to compute expected flood losses under both scenario-based and probabilistic approaches.

3.1 Hazard and Depth

Flood hazard refers to the intensity and spatial extent of flooding, typically expressed as inundation depth for different return periods. In this study, hazard is represented as a function that maps location and return period to predicted water depth.

3.1.1 Mathematical Definition of Flood Hazard

Flood hazard is modeled as a spatially varying flood depth field:

$$h : \Omega \times \mathcal{T} \to \mathbb{R}_{\geq 0}, \quad (x, y, T) \mapsto h(x, y, T) \tag{3.1}$$

where:

- $\Omega \subset \mathbb{R}^2$ is the spatial domain (study area),
- $\mathcal{T} = \{T_1, T_2, \ldots, T_K\}$ is the set of modeled return periods (e.g., 10, 100, 500 years),
- $h(x, y, T)$ is the flood depth at location (x, y) under return period T.

The return period T expresses the average time between flood events of that magnitude, while the annual exceedance probability (AEP) is its inverse:

$$p = \frac{1}{T} \tag{3.2}$$

Flood hazard maps $h(x, y, T)$ are derived from hydrological and hydraulic models that simulate flood dynamics based on rainfall, runoff, topography, and river systems. In practice, the continuous function $h(x, y, T)$ is approximated by raster arrays $H_{i,j,k}$, where i, j index the spatial grid and k indexes the return period.

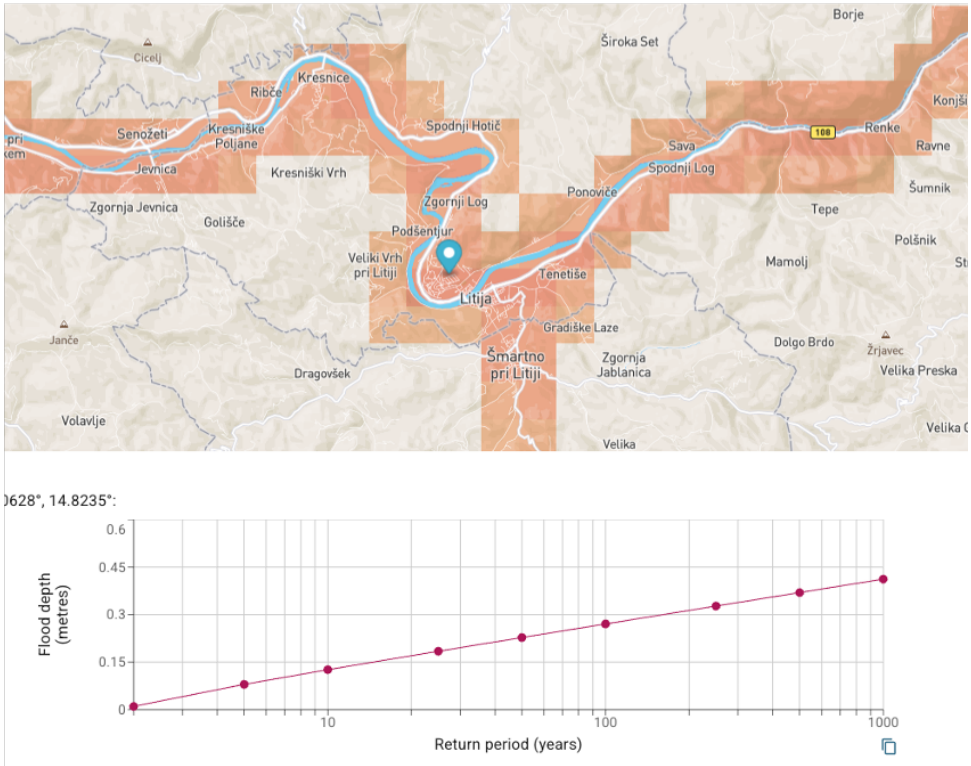

Figure 3.1: Flood hazard illustration for a location in Slovenia (Litija). The top panel shows spatial coverage of flood modeling; the bottom panel shows flood depth $h(x, y, T)$ versus return period T.

3.1.2 Hazard Data Sources and Modeling Approach

Two complementary hazard datasets are used in this study:

- **WRI Aqueduct River Flood Model:** A global flood hazard dataset developed by the World Resources Institute. It provides flood depth estimates at 1 km resolution for return periods from 2 to 1000 years, using simulations based on historical river discharge, terrain data (e.g., MERIT DEM), and land surface parameters. Critically, this model incorporates climate scenarios through Representative Concentration Pathways (RCPs), including:
  - **RCP4.5**: A moderate emissions pathway, stabilizing radiative forcing at 4.5 W/m^2 by 2100.
  - **RCP8.5**: A high-emissions scenario, corresponding to >4°C warming by 2100.

  Outputs are provided in GeoTIFF raster format (EPSG:4326).

- **Integralna karta globin (IKG):** A set of high-resolution flood hazard maps produced by Slovenian civil protection authorities. These are based on 1D/2D hydraulic modeling for return periods of 10, 100, and 500 years. The original data are provided as vector polygons categorized into three flood depth bands:
  - GM – Minor inundation (up to 0.5 m),
  - GS – Significant inundation (up to 1.5 m),
  - GB – Severe inundation (up to 2.5 m).

IKG models are locally calibrated using historical flood events and topographic surveys. Vector inputs are rasterized to a 5 m grid in EPSG:3794 and harmonized with the other spatial layers.

Table 3.1: Comparison of flood hazard models used in the study.

Feature	**WRI Aqueduct River Flood**	**Integralna karta globin (IKG)**
Spatial resolution	∼1 km	5 m
Coverage	Global	18 regions in Slovenia
Climate scenarios	RCP-aware (RCP4.5, RCP8.5)	Not scenario-aware
Depth type	Continuous (in meters)	Discrete classes (0.5, 1.5, 2.5 m)
Return periods	2 to 1000 years	10, 100, 500 years
File format	Raster (GeoTIFF)	Vector (shapefile, GPKG)
Coordinate system	EPSG:4326	EPSG:3794 (transformed)

To enable direct comparison and model evaluation, all flood depth maps are resampled to a common 5 m grid and spatial reference system. These harmonized hazard layers serve as inputs to the vulnerability and risk modeling framework described in the following sections.

3.2 Vulnerability and Damage Function

Flood vulnerability quantifies the degree to which an asset is susceptible to damage when exposed to flooding. In quantitative modeling, this is typically represented using **depth–damage functions** (DDFs), which estimate the fraction of asset value lost as a function of water depth.

3.2.1 Mathematical Definition of Flood Vulnerability

A depth–damage function $D(h)$ maps flood depth h (in meters) to the expected relative damage $D(h) \in [0, 1]$:

$$D : \mathbb{R}_{\geq 0} \to [0, 1], \quad h \mapsto D(h) \tag{3.3}$$

These functions are typically monotonically increasing and piecewise continuous. They begin at a damage threshold (e.g., 0.1–0.2 m) and increase until they saturate near the total loss level (typically 70–90% of asset value), depending on construction type and use.

3.2.2 Vulnerability Data Source

This thesis uses a set of standardized depth–damage functions developed by the European Commission Joint Research Centre (JRC) and published by Huizinga et al. [7]. These functions are among the most widely used tools for flood damage estimation in Europe.

The DDFs are stratified by:

- **Asset type** (e.g., residential, commercial, industrial),
- **Damage component** (structure, contents, or total),
- **Construction context** (regional economic and building typologies).

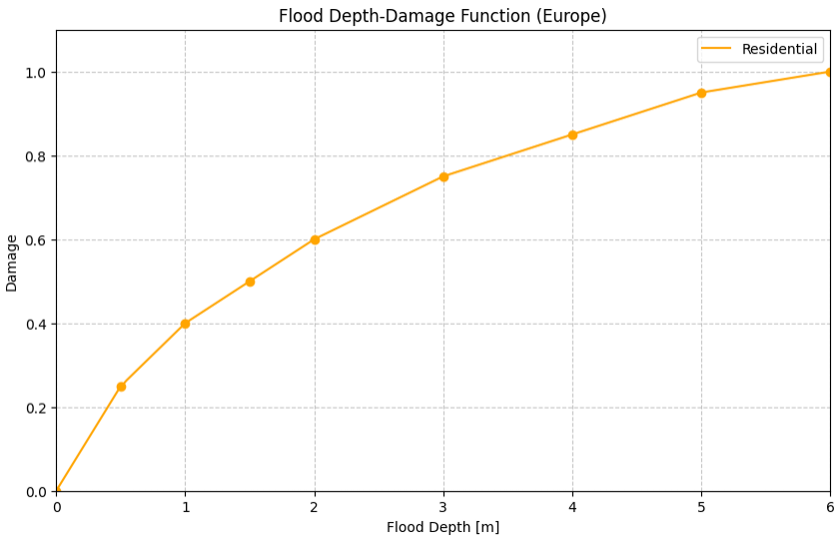

Figure 3.2: Standardized depth–damage function for residential buildings in Europe, based on Huizinga et al. [7]. The curve indicates increasing expected damage with flood depth, saturating near total loss.

For each property in the Slovenian flood dataset, a vulnerability curve was selected based on asset classification. All damage functions are calibrated relative to building replacement value (BRV), which serves as the basis for expressing relative damage.

3.2.3 Validation Dataset

Observed flood damage data were obtained from the Administration for Civil Protection and Disaster Relief of Slovenia. This dataset includes:

- Georeferenced flood-affected locations,
- Estimated or recorded flood depths,
- Reported monetary damage in EUR,
- Metadata on property type and use (where available).

Reported values were normalized by estimated replacement value, allowing comparison with DDF predictions. Where metadata was missing, building type was inferred using cadastral layers or estimated from available zoning attributes.

3.2.4 Assumptions and Limitations

- **Transferability:** The JRC functions are designed for generalized European use and may not perfectly reflect local construction practices in Slovenia.
- **Uncertainty:** The functions represent expected values and do not account for uncertainty or variance in actual damage.
- **Data completeness:** Some events in the validation dataset lack full metadata or have coarse depth estimates, limiting precision.

Despite these limitations, the global vulnerability curves offer a consistent and replicable benchmark for testing flood risk models and have been widely used in EU-level assessments.

3.3 Flood Risk Modeling

Flood risk refers to the expected monetary loss due to flooding, calculated by combining the hazard (flood depth), vulnerability (damage function), and exposure (asset value). This section presents the mathematical framework used in this thesis, distinguishing between losses for a specific return period and aggregated probabilistic estimates via **Expected Annual Damage (EAD)**.

3.3.1 Risk for a Single Return Period

For a given return period T, the expected loss at location $(x, y) \in \Omega$ is given by:

$$R(x, y; T) = D(h(x, y, T)) \cdot E(x, y) \tag{3.4}$$

where:

- $h(x, y, T)$ is the flood depth at location (x, y) under return period T,
- $D(h)$ is the depth–damage function,
- $E(x, y)$ is the monetary value of the asset at (x, y).

This formulation corresponds to a scenario-based risk estimate, conditional on the occurrence of a specific flood event. For example, if a building is worth €100,000, and the hazard model predicts $h = 0.31\,\text{m}$ under RP100, and $D(0.31) = 0.17$, the expected damage is:

$$R(x, y; 100) = 0.17 \cdot 100{,}000\,€ = 17{,}000\,€$$

To estimate total loss over the study area Ω, the pointwise losses are integrated:

$$R_{\text{total}}(T) = \int_{\Omega} D(h(x, y, T)) \cdot E(x, y)\, dx\, dy \tag{3.5}$$

For gridded data, this integral is approximated by a discrete sum:

$$R_{\text{total}}(T_k) \approx \sum_{i=1}^{N_x} \sum_{j=1}^{N_y} D(H_{i,j,k}) \cdot E_{i,j} \tag{3.6}$$

3.3.2 Expected Annual Damage (EAD)

While single-return-period losses are useful, financial and regulatory applications often require a risk indicator that incorporates the full probability distribution of flood events. The **Expected Annual Damage (EAD)** is such a measure: it aggregates flood risk across a range of return periods using their corresponding annual exceedance probabilities (AEPs).

Let $T_k \in \mathcal{T}$ denote the set of modeled return periods, with associated exceedance probabilities $p_k = 1/T_k$. The EAD at location (x, y) is then:

$$\text{EAD}(x, y) = \int_0^1 R(x, y; p)\, dp = \int_0^1 D(h(p)) \cdot E(x, y)\, dp \tag{3.7}$$

In practice, this integral is approximated via trapezoidal integration over a discrete set of return periods:

$$\mathrm{EAD}(x, y) \approx \sum_{k=1}^{K-1} \frac{p_k - p_{k+1}}{2} \cdot [R(x, y; T_k) + R(x, y; T_{k+1})]$$

This formulation captures both frequent and rare flood events, producing an annualized estimate of risk suitable for insurance, portfolio analysis, and regulatory reporting.

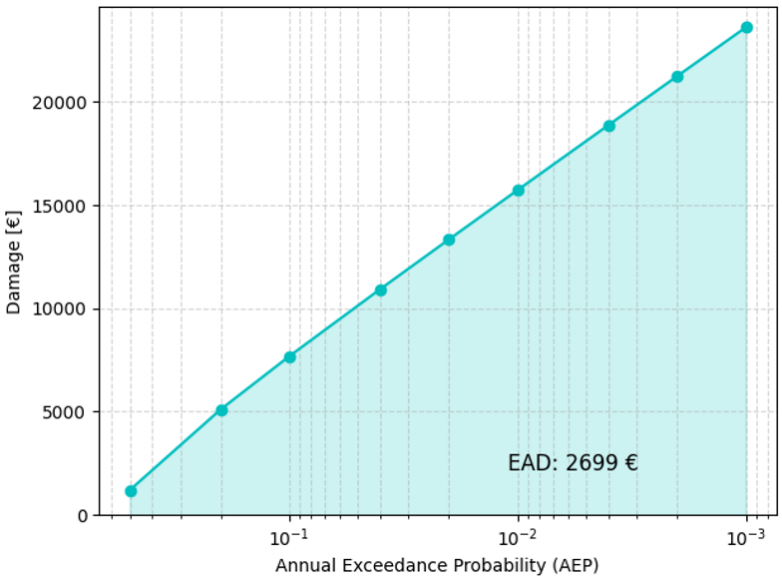

Figure 3.3: Visualization of Expected Annual Damage (EAD). The curve plots flood damage as a function of annual exceedance probability; the shaded area under the curve represents EAD.

3.3.3 Exposure Data

Exposure refers to the monetary value of physical assets affected by flooding. In this thesis, exposure is represented in a simplified form using fixed nominal values. This was done to focus the analysis on hazard and vulnerability model performance, rather than on variability in asset valuation.

In the illustrative examples and validation scenarios, a synthetic property value of €100,000 was used. This allowed flood risk to be modeled consistently across different hazard and vulnerability combinations.

Importantly, the validation analysis was conducted in **relative terms** — comparing observed and predicted *fractional damage* (i.e., damage as a share of estimated property value), rather than in absolute monetary losses. This avoids the need for detailed cadastral or tax registry data and reduces dependence on asset-specific exposure assumptions.

While this approach abstracts away exposure uncertainty, it enables transparent model comparison. Integration of more granular exposure data (e.g. official appraisal values) is identified as an area for future work.

3.3.4 Use Cases and Applications

The risk modeling framework presented here supports several key applications:

- **Model comparison:** Evaluate differences in hazard predictions between global and local models.
- **Damage model validation:** Test how well global DDFs explain observed flood impacts in Slovenia.
- **EAD estimation:** Produce standardized annualized loss estimates for reporting and disclosure.
- **Sensitivity analysis:** Assess the impact of spatial resolution, proximity to water bodies, and hazard assumptions.
- **Modular integration:** Enable ingestion of local models into the Physrisk platform for institutional ESG workflows.

Chapter 4

Model Integration and Analysis Framework

Accurate flood risk assessment requires the integration of diverse data sources and computational tools to simulate hazard intensity, estimate potential damages, and evaluate financial exposure. In this chapter, we present the technical framework developed to combine heterogeneous flood hazard data—particularly local Slovenian models—into a unified system suitable for analysis and comparison within the Physrisk platform.

The focus lies on harmonizing the *Integralna karta globin* (IKG) flood maps, which vary in structure, format, and coordinate reference systems, and transforming them into a consistent, machine-readable representation. These models are rasterized, standardized by return period, and encoded in the cloud-optimized Zarr format. The processed datasets are then onboarded into the modular Physrisk platform, which separates hazard, vulnerability, and financial components and allows for flexible model evaluation.

To facilitate empirical comparison, the framework also includes a toolset that enables side-by-side querying and batch evaluation of both global and local models at precise geolocations, aligned with the validation dataset. This infrastructure supports all subsequent experimental analysis presented in Chapter 5 and lays the groundwork for scalable, reproducible climate risk assessments.

The chapter is organized as follows:

- Section 4.1 presents the overall system architecture and the interaction between components.
- Section 4.2 describes the pipeline for transforming and combining IKG models across multiple Slovenian regions.
- Section 4.3 explains how the standardized hazard data are integrated into the Physrisk platform.
- Section 4.4 outlines the model comparison toolset used to evaluate prediction accuracy across models.

4.1 Architectural Diagram

The integration of global and local flood hazard models into a unified analysis platform requires a modular, extensible architecture that supports heterogeneous input formats, scenario-aware modeling, and scalable evaluation. This section presents the architecture developed for this purpose, highlighting its components, data flow, and interoperability within the *Physrisk* framework.

4.1.1 System Overview

Figure **??** illustrates the architecture of the flood hazard model integration pipeline, based on the open-source *OS-Climate Physrisk* ecosystem. The process begins with raw flood hazard data, which may originate from global sources such as the WRI Aqueduct or TU Delft flood models, or from local sources such as the Integralna karta globin (IKG) used in Slovenia. These datasets are onboarded into the `os-risk-hazard` module, a Python-based processing library designed to standardize, harmonize, and prepare hazard indicators for risk evaluation.

The onboarded models are then encoded in Zarr format and stored in an AWS S3-compatible object storage. The Physrisk framework retrieves hazard indicators from this backend and combines them with vulnerability and exposure components to compute impact distributions. The resulting outputs can be accessed via the Physrisk API or used directly for ESG reporting and financial risk analysis through the `os-risk-financial` module.

4.1.2 Data Flow and Model Interoperability

Each flood model—whether global or local—is processed through the same onboarding interface, allowing standardization of outputs while preserving return-period-specific intensity information. The hazard models are structured into three-dimensional Zarr arrays indexed by latitude, longitude, and return period. This enables consistent querying and evaluation across models with different spatial resolutions or geographic coverage.

Once onboarded, all hazard layers become interchangeable within the Physrisk pipeline. Vulnerability models (e.g., depth–damage functions) and exposure datasets (e.g., building values or asset locations) can then be used in combination with any hazard source. This interoperability facilitates both comparative experiments and compound risk evaluations, where multiple models are used to bound uncertainty or stress-test assumptions.

4.1.3 Integration of the IKG Model

The locally developed IKG model, originally provided in shapefile and GeoPackage vector formats, required a dedicated pre-processing stage to align with the Physrisk architecture. This process involved:

- Transforming coordinate systems (from EPSG:3794 to EPSG:4326),
- Rasterizing vector layers into a fixed-resolution 5x5m grid,
- Assigning discrete flood depth values to each return period (RP10, RP100, RP500),
- Converting the raster layers into Zarr format for chunked cloud storage.

The unified model was then uploaded to AWS S3 and integrated into the Physrisk hazard module alongside WRI and TU Delft models. This enabled direct comparison and batch evaluation of global versus local model predictions, described further in Section 4.4.

4.1.4 Hazard Onboarding Pipeline

The transformation from raw IKG data into a cloud-optimized flood hazard model is summarized in Figure **??**. The pipeline consists of the following steps:

1. **Rasterization:** Converting input vector data into gridded raster tiles using a 5x5m resolution grid.

2. **Zarrify:** Chunking the raster into 1000x1000 tiles and encoding into Zarr format.
3. **Upload:** Transferring the data to AWS S3 for use within Physrisk.

The final onboarded dataset can be queried via the Physrisk API or accessed directly for integration with ESG tools or batch model evaluation.

4.2 Pipeline for Combining Heterogeneous IKG Models

The *Integralna karta globin* (IKG) flood hazard datasets are developed for individual municipalities or regions in Slovenia, each based on local hydrological and hydraulic analyses. These datasets differ in file structure, geospatial reference systems, data quality, and encoding of flood depths. To integrate these into a unified flood hazard model suitable for comparative evaluation and platform onboarding, a standardized processing pipeline was developed.

4.2.1 Data Sources and Heterogeneity

The raw IKG datasets were obtained from multiple engineering consultancies (*Inženiring za vode d.o.o., Urbania d.o.o., iS Projekt d.o.o., IZVOR d.o.o., DHD d.o.o.*), each responsible for flood modeling in a subset of areas classified as "Območja pomembnega vpliva" (Areas of Significant Flood Risk) by Slovenian water authorities.

Each local model typically includes:

- GIS vector data (shapefiles, GeoPackages) representing inundation zones,
- Attribute data encoding flood extent and classes for selected return periods (e.g., 10-, 100-, and 500-year events),
- Region-specific coordinate systems (typically EPSG:3794, Slovenia National Grid).

In total, data from 18 areas of importance were collected, with each area providing flood scenarios for three return periods. This resulted in 54 distinct layers that needed to be harmonized.

4.2.2 Standardization and Processing Workflow

To unify the IKG datasets into a single model, the following pipeline was developed and implemented in Python using open-source geospatial processing libraries (GDAL, Fiona, Rasterio, xarray):

1. **Coordinate Transformation:** All source files were transformed from EPSG:3794 to WGS84 (EPSG:4326) to ensure compatibility with global models and standardized grid alignment.
2. **Rasterization:** Vector layers were rasterized into a regular grid with a resolution of 5 meters (5x5m), which approximates the native resolution of the original hydrological models. A fixed flood depth value was assigned to each flood class:
    - GM (minor flooding): 0.5 m
    - GS (significant flooding): 1.5 m
    - GB (severe flooding): 2.5 m

3. **Return Period Alignment:** Each rasterized layer was tagged with its associated return period (RP10, RP100, RP500). For each area, three separate rasters were produced and stacked by return period.

4. **Tiling and Merging:** The processed rasters from all areas were mosaicked into a common spatial grid and stored as a 3D array indexed by latitude, longitude, and return period. Conflicts in overlapping regions (e.g., model boundaries) were resolved by assigning priority to higher-confidence datasets based on metadata quality.

4.2.3 Unified Flood Hazard Product

The final result of this pipeline is a unified flood hazard model for Slovenia covering all 18 areas of importance. This model:

- Captures flood depth estimates at high spatial resolution for three return periods,
- Standardizes input data from diverse local models into a single harmonized format,
- Enables consistent comparison with global models such as WRI Aqueduct within the same framework.

Figure **??** provides a schematic representation of the transformation process, from raw shapefiles to a standardized hazard product.

This unified model was subsequently converted to Zarr format and onboarded into the Physrisk platform, as described in Section 4.3.

4.3 Physrisk Framework – Model Onboarding

Once the local IKG flood models were unified into a standardized and spatially consistent format, the next step involved integrating them into the modular *Physrisk* framework. Physrisk is part of the broader OS-Climate open-source ecosystem and is designed to facilitate scalable, scenario-based physical climate risk analysis. The framework separates hazard, vulnerability, and financial components and enables flexible model composition, comparison, and evaluation.

4.3.1 Physrisk Architecture and Modules

The Physrisk architecture is composed of several interoperable modules:

- **OS-risk-hazard**: responsible for ingesting and formatting hazard data sets (e.g., flood depth by return period) into a common format.
- **OS-risk-physrisk**: computes impact distributions by convolving hazard data with vulnerability functions and exposure information.
- **OS-risk-financial**: translates physical impacts into financial metrics such as loss of asset value or disruption to operations.
- **Physrisk API**: exposes model outputs through a standardized programmatic interface.

The onboarding of hazard models occurs within the `os-risk-hazard` component and requires the hazard data to conform to a Zarr-based storage structure optimized for parallel access and spatial querying.

4.3.2 Zarr Format and Indexing

The Zarr format is a cloud-native format for storing large multi-dimensional arrays. It was chosen for hazard model storage due to its ability to:

- Efficiently compress and store gridded data in chunks (e.g., 1000×1000 tiles),
- Allow selective download of spatial subsets,
- Support multi-indexing (e.g., latitude, longitude, return period).

For the unified IKG model, the hazard data were encoded into a 3-dimensional Zarr array with the following dimensions:

- i: Latitude (y-axis),
- j: Longitude (x-axis),
- k: Return period index, corresponding to RP10, RP100, and RP500.

Each tile was stored as a compressed chunk and uploaded to an AWS S3-compatible bucket accessible to all Physrisk modules.

4.3.3 Onboarding Procedure

The onboarding process was implemented using the Physrisk hazard onboarding tool, a Python-based utility designed for structured hazard model integration. The steps were as follows:

1. **Prepare Metadata:** Define the return periods, CRS (EPSG:4326), spatial resolution, and transformation matrix for the IKG model.
2. **Convert to Zarr:** Use the onboarding tool to encode the processed GeoTIFF files into a Zarr store, assigning appropriate chunk sizes (e.g., 1000×1000 pixels).
3. **Upload to Cloud Storage:** Transfer the Zarr directory to a cloud-accessible S3 bucket used by the OS-Climate infrastructure.
4. **Register the Model:** Add a model reference entry into the hazard configuration of Physrisk, enabling downstream access via API and batch processors.

4.3.4 Integration with Other Models

With the IKG model onboarded, it became available in the same infrastructure alongside global flood hazard layers, including:

- WRI Aqueduct River Flood (global, 1 km resolution),
- TU Delft River Flood (Pan-European, 100 m resolution).

This interoperability allows any of the onboarded hazard layers to be used with common vulnerability functions and exposure datasets, enabling direct comparisons, uncertainty bounding, and ensemble evaluations. The IKG model is now fully accessible via the Physrisk API and can be used for property-level flood risk analysis in Slovenia under a unified modeling interface.

4.4 Comparison Toolset

To evaluate and compare the predictive performance of flood hazard models, a model-agnostic comparison toolset was developed and integrated into the Physrisk framework. This toolset enables side-by-side querying of multiple hazard sources at specific geolocations and return periods, aligned with the reference flood observations used for validation in Chapter 5.

4.4.1 Purpose and Design Principles

The comparison toolset was designed with the following goals in mind:

- **Model-agnostic access:** Allow querying of different onboarded hazard models (e.g., IKG, WRI Aqueduct) through a unified interface.
- **Return-period specificity:** Ensure comparability by extracting flood depth predictions at the same return period (e.g., 100-year flood).
- **Location-accurate evaluation:** Retrieve model outputs at high spatial precision to match reported flood locations.
- **Batch processing support:** Enable evaluation across hundreds of observations using efficient geospatial access patterns.

4.4.2 Interface and Functionality

The toolset consists of a lightweight Python interface built on top of the Physrisk hazard API. Given a set of coordinates (latitude, longitude) and a desired return period, the tool performs the following steps:

1. **Coordinate Transformation:** Convert input coordinates into the grid index of the onboarded Zarr array using affine transformations defined in the model metadata.
2. **Chunk Identification and Retrieval:** Locate the Zarr chunk containing the requested point and load the corresponding depth value for the specified return period.
3. **Output Formatting:** Return the flood depth value, return period, and model metadata for downstream use in regression or classification analysis.

When applied in batch mode, the tool leverages parallel reads and lazy loading to evaluate large datasets (e.g., the full validation set of flood reports) with minimal memory overhead.

4.4.3 Supported Models and Inputs

The toolset was validated for use with the following onboarded hazard models:

- **SI IKG (Unified Model)**: High-resolution local flood model rasterized at 5x5 m, covering 18 Slovenian areas of importance, with RP10, RP100, and RP500 layers.
- **WRI Aqueduct Flood Hazard**: Global model with 1 km resolution, scenario-aware, supporting multiple return periods from RP2 to RP1000.

The input to the toolset is a tabular dataset containing:

- Latitude and longitude of each observation,
- Desired return period (e.g., RP100),
- Optional fields: timestamp, municipality, reported depth, reported damage.

4.4.4 Integration with Validation Workflow

The comparison outputs are stored in a structured table, allowing for direct computation of model performance metrics such as:

- **Regression error:** difference between predicted and observed flood depths,
- **Classification accuracy:** match rate within flood depth categories (e.g., <0.5 m, 0.5–1.5 m, >1.5 m),
- **Bias analysis:** mean and median error segmented by proximity to water, terrain type, or municipality.

These outputs form the quantitative basis for the experimental results presented in Chapter 5.

4.4.5 Reproducibility and Extensibility

The toolset was developed as a standalone module and version-controlled as part of the project's repository. This design ensures that future updates to hazard models (e.g., inclusion of climate scenario variants or additional regions) can be incorporated without changes to the validation logic.

Furthermore, the modularity of the toolset supports extension to:

- Additional hazard types (e.g., pluvial flooding),
- Vulnerability assessments (via model chaining),
- Scenario-based risk projections (by switching hazard layers).

Thus, the comparison toolset serves not only as a core utility for this thesis, but also as a scalable evaluation framework for broader applications in physical climate risk modeling.

Chapter 5

Experimental Design and Evaluation

TBD

5.1 Experimental Goals

TBD

5.2 Experimental Setup

TBD

5.3 Experimental Results: Flood Depth

TBD

5.4 Experimental Results: Damage Functions

TBD

5.5 Analysis of Impact on Total Exposed Property Value

TBD

Chapter 6

Conclusions & Discussion

6.1 Conclusion & Interpretation of Results

TBD

6.2 Limitations of This Work

TBD

6.3 Future Work

TBD

Appendix A

Proofs of Theorems

A.1 Proof of the Pythagorean Theorem

Let us prove the Pythagorean Theorem from page **??**.

Proof. This proof is based on the proportionality of the sides of two similar triangles, that is, upon the fact that the ratio of any two corresponding sides of similar triangles is the same regardless of the size of the triangles.

Let ABC represent a right triangle, with the right angle located at C, as shown in Figure A.1. We draw the altitude from point C, and call H its intersection with the hypotenuse AB. Point H divides the length of the hypotenuse into two parts.

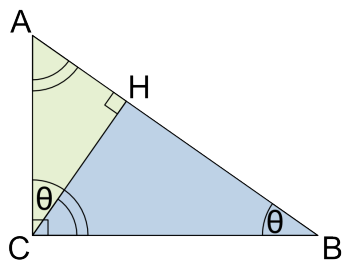

Figure A.1: Similar triangles used in the proof of the Pythagorean theorem.

The new triangle ACH is similar to triangle ABC, because they both have a right angle (by definition of the altitude), and they share the angle at A, meaning that the third angle will be the same in both triangles as well, marked as θ in Figure A.1. By a similar reasoning, the triangle CBH is also similar to ABC.

Similarity of the triangles leads to the equality of ratios of corresponding sides:

$$\frac{BC}{AB} = \frac{BH}{BC} \text{ and } \frac{AC}{AB} = \frac{AH}{AC}. \tag{A.1}$$

The first result equates $\cos\theta$ and the second result equates $\sin\theta$.

These ratios can be written as:

$$BC^2 = AB \times BH \text{ and } AC^2 = AB \times AH. \tag{A.2}$$

Summing these two equalities, we obtain:

$$BC^2 + AC^2 = AB \times BH + AB \times AH = AB \times (AH + BH) = AB^2, \tag{A.3}$$

which, tidying up, is the Pythagorean theorem:

$$BC^2 + AC^2 = AB^2. \tag{A.4}$$

□

References

[1] IPCC, *Climate change 2021: The physical science basis. contribution of working group i to the sixth assessment report of the intergovernmental panel on climate change*, 2021. [Online]. Available: https://www.ipcc.ch/report/ar6/wg1/.

[2] European Environment Agency, "Urban adaptation in europe: How cities and towns respond to climate change," European Environment Agency, Tech. Rep., 2020. [Online]. Available: https://www.eea.europa.eu/publications/urban-adaptation-in-europe.

[3] European Central Bank, "Guide on climate-related and environmental risks: Supervisory expectations relating to risk management and disclosure," European Central Bank, Tech. Rep., Nov. 2020. [Online]. Available: https://www.bankingsupervision.europa.eu/ecb/pub/pdf/ssm.202011finalguideonclimate-relatedandenvironmentalrisks~58en.pdf.

[4] Task Force on Climate-related Financial Disclosures, "Final report: Recommendations of the task force on climate-related financial disclosures," Financial Stability Board, Tech. Rep., Jun. 2017. [Online]. Available: https://www.fsb-tcfd.org/publications/final-recommendations-report/.

[5] J. Moorhouse, F. Gallo, M. Bouchaala, and D. Ferri, "Physical Climate Risk Methodology," Feb. 2024, Version 1.0.

[6] J. Woetzel, D. Pinner, and H. Samandari, "Climate risk and response: Physical hazards and socioeconomic impacts," 2020.

[7] J. Huizinga, H. d. Moel, and W. Szewczyk, "Global flood depth-damage functions: Methodology and the database with guidelines," *JRC Research Reports*, Apr. 2017, Number: JRC105688, Publisher: Joint Research Centre.

Bibliography

Publications Related to the Thesis

All publications related to the thesis should be referenced in the text.

Other Publications (optional)

...

Biography

The author of this thesis ...

Index

Pythagorean theorem
    proof, 25